Practical Digital Systems Design

Homework 4:

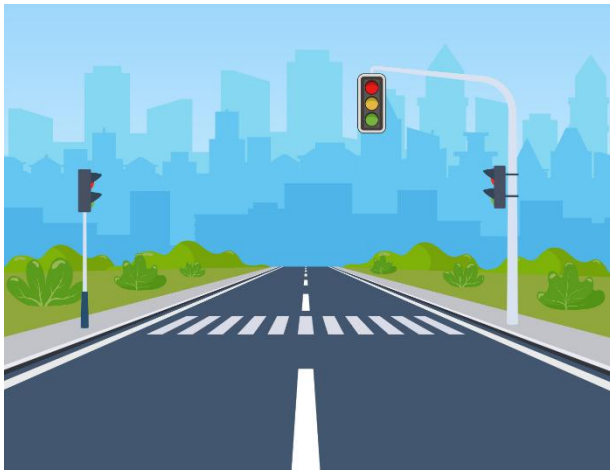
Traffic Light Control Circuit

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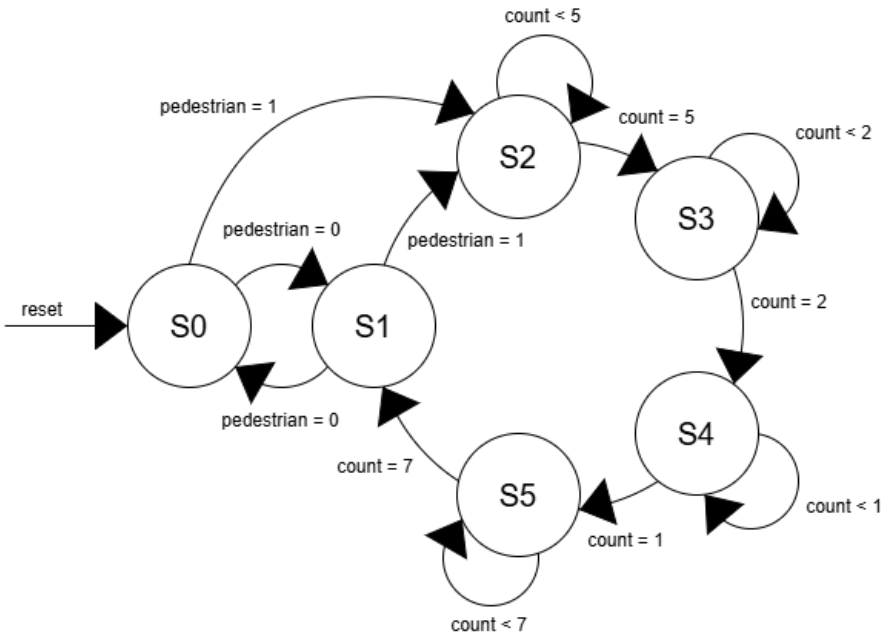
➢ Design Principle

By default, the main road light flashes yellow while the pedestrian signal flashes red. When the pedestrian button is pressed, this flashing pattern stops and a timed sequence begins: the vehicle light turns green while the pedestrian signal remains red, then the main signal shifts from green to yellow and finally to red, after which the pedestrian signal switches to green. When the crossing interval ends, both signals returns to the original flashing pattern.



Define the main road traffic light as Light1 and the pedestrian traffic light as Light2.

State	Light1	Light2	Duration
S0	Yellow (010)	OFF (00)	1s
S1	OFF (000)	Red (10)	1s
S2	Green (001)	Red (10)	6s
S3	Yellow (010)	Red (10)	3s
S4	Red (100)	Red (10)	2s
S5	Red (100)	Green (01)	8s



➢ Circuit Structure

I. Synchronous Counter

There are three control signals in the counter: reset, clear, and enable. The reset signal resets the entire system, including the counter. The clear signal is asserted at the end of each state to return the counter to zero, ensuring that each new state begins timing from a starting point. The enable signal controls whether the counter is allowed to increment. The synchronous counter increases its value on every rising edge of the clock when it is not being reset or cleared and when enable is asserted, providing the timing reference that determines how long each state should remain active.



II. Traffic Light Controller

The traffic-light controller is implemented as a six-state finite state machine encoded with three bits. In normal operation, it alternates between two states to create the flashing yellow pattern for the main road and the flashing red pattern for pedestrians. When the pedestrian button is pressed, the controller exits this flashing sequence and progresses through a timed series of states: the vehicle light first turns green, then transitions through yellow and finally to red, giving cars sufficient time to clear the intersection before switching the pedestrian signal to green. Each state's duration is determined by the counter, allowing the controller to precisely control how long each signal remains active. Once the pedestrian crossing interval ends, the controller returns to the original flashing pattern. The output logic assigns the appropriate bit patterns to light1 and light2, where light1 uses three bits to represent red (100), yellow (010), green (001), or off (000), and light2 uses two bits to represent red (10), green (01), or off (00).



➢ Synthesis Flow

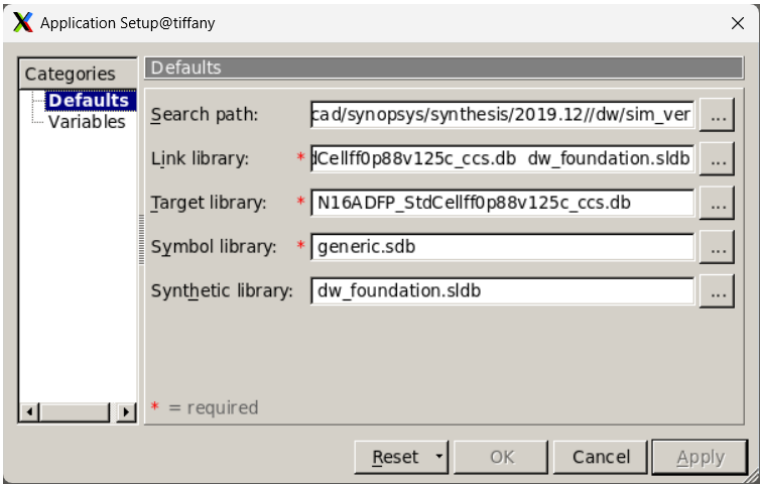
1. Upload the **.synopsys_dc.setup** file to the RCA/ directory via FileZilla.
2. In the Xshell terminal, enter **ls -al** to verify that the correct configuration file is included.

```
[B123015021@tiffany ~/HW4]$ ls -a
.  ..  N16ADFP_StdCell.v  .synopsys_dc.setup  tb_traffic_light.v  traffic_light.v
```

3. In the Xshell terminal, enter **ncverilog tb_traffic_light.v +access+r** to compile and simulate traffic_light.v, and verify that the simulation runs correctly.



4. In the Xshell terminal, enter **dv &** and wait for Design Vision to launch.
5. In the Toolbox, select **File → Setup** and verify that the library contents are correctly configured.



6. Load the top.v file to be synthesized, and verify that the message **"Presto compilation completed successfully"** is displayed.

```
Presto compilation completed successfully.
Current design is now '/home/UGP/B123015021/HW4/traffic_light.db:traffic_light'
Loaded 2 designs.
Current design is 'traffic_light'.
```

7. Verify that the hierarchy is set to the top module. Otherwise, enter the command **current_design traffic_light** in the command line.

```
design_vision> current_design traffic_light
Current design is 'traffic_light'.
{traffic_light}
```

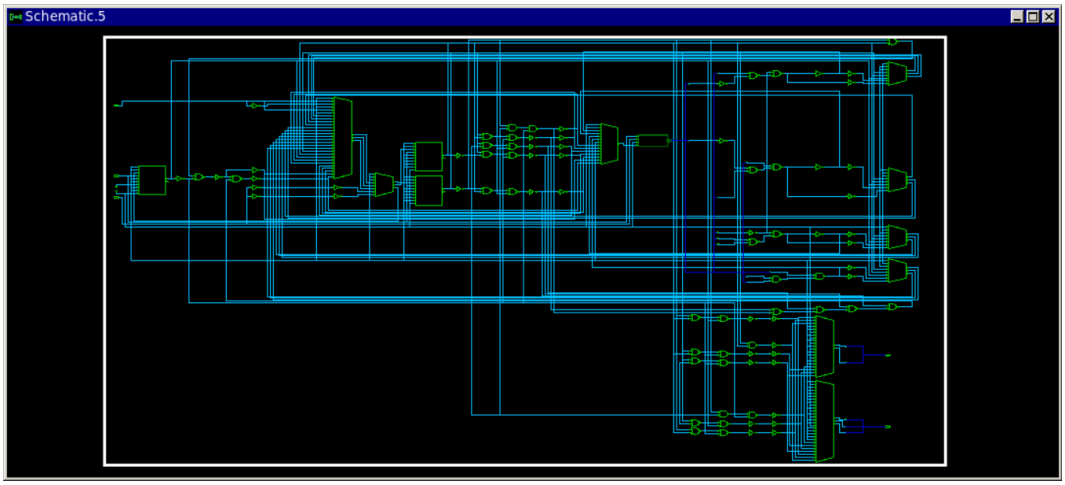
8. In the command line, enter: **set_fix_multiple_port_nets -all -buffer_constants**

```
design_vision> set_fix_multiple_port_nets -all -buffer_constants
1
```

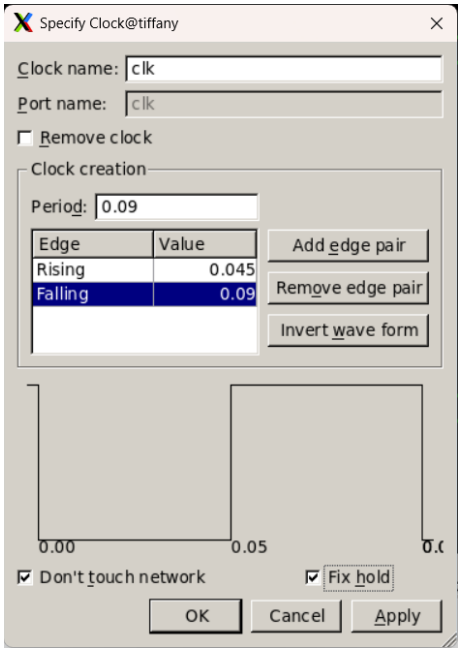
9. View the Symbol View.



10. View the Schematic View.



11. Highlight the clock in the Symbol View, then go to **Attributes → Specify Clock** in the toolbox, and set the clock period as well as the rise and fall times.



12. In the Toolbox, select **Design → Compile Design**, and verify that the message **"Optimization Complete"** is displayed.

```
Optimization Complete
---------------------
1
```

13. Using **echo $target_library** allows you to check whether the current library setting is correct.

```
design_vision> echo $target_library
N16ADFP_StdCellff0p88v125c_ccs.db
```

14. In the Toolbox, select **Design → Report Area** to view the area report.

```
****************************************
Report : area
Design : traffic_light
Version: Q-2019.12
Date   : Tue Nov 25 17:14:35 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                          15
Number of nets:                           84
Number of cells:                          75
Number of combinational cells:            68
Number of sequential cells:                6
Number of macros/black boxes:              0
Number of buf/inv:                        21
Number of references:                     27

Combinational area:                17.314560
Buf/Inv area:                       3.265920
Noncombinational area:              5.598720
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   22.913281
Total area:                 undefined

***** End Of Report *****
```

15. In the Toolbox, select **Design → Report Power** to view the power report.

```
  Cell Internal Power  = 310.1151 uW   (80%)
  Net Switching Power  =  77.6092 uW   (20%)
                         ---------
Total Dynamic Power    = 387.7244 uW  (100%)

Cell Leakage Power     =  31.1423 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000              0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000              0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000              0.0000             0.0000  (   0.00%)
register           0.2394        2.1280e-02          1.0021e+04            0.2707  (  64.62%)
sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
combinational  7.0733e-02        5.6329e-02          2.1121e+04            0.1482  (  35.38%)
--------------------------------------------------------------------------------------------------
Total              0.3101 mW     7.7609e-02 mW     3.1142e+04 nW          0.4189 mW

***** End Of Report *****
```

16. In the Toolbox, select **Timing → Report Timing Path** to view the delay time from the path's starting point to each node.
    i. max

```
Point                                                   Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                   0.05       0.05
clock network delay (ideal)                             0.00       0.05
cs_reg[0]/CP (DFQD2BWP16P90LVT)                         0.00       0.05 r
cs_reg[0]/Q (DFQD2BWP16P90LVT)                          0.03       0.07 f
U69/ZN (INVD1BWP16P90LVT)                               0.01       0.08 r
U54/Z (AN3D1BWP16P90LVT)                                0.01       0.09 r
U55/ZN (ND2D2BWP16P90LVT)                               0.01       0.10 f
U60/ZN (ND2D2BWP16P90LVT)                               0.00       0.10 r
U45/ZN (ND2D2BWP16P90LVT)                               0.01       0.11 f
U51/ZN (AOAI211D1BWP16P90LVT)                           0.01       0.12 r
U83/Z (OA21D1BWP16P90LVT)                               0.01       0.13 r
cs_reg[0]/D (DFQD2BWP16P90LVT)                          0.00       0.13 r
data arrival time                                                  0.13

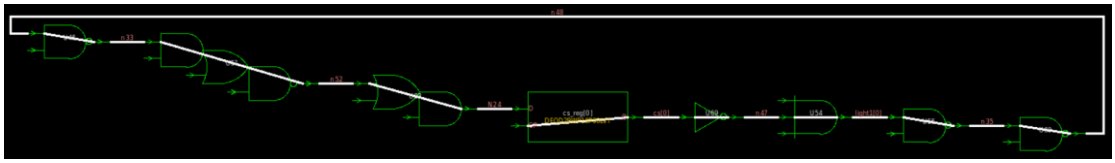
clock clk (rise edge)                                   0.14       0.14
clock network delay (ideal)                             0.00       0.14
cs_reg[0]/CP (DFQD2BWP16P90LVT)                         0.00       0.14 r
library setup time                                     -0.01       0.13
data required time                                                 0.13
-----------------------------------------------------------------------
data required time                                                 0.13
data arrival time                                                 -0.13
-----------------------------------------------------------------------
slack (MET)                                                        0.00
```

ii. min

```
Point                                                   Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                   0.05       0.05
clock network delay (ideal)                             0.00       0.05
counter/count_reg[0]/CP (DFQD2BWP16P90LVT)              0.00       0.05 r
counter/count_reg[0]/Q (DFQD2BWP16P90LVT)               0.03       0.07 r
counter/U14/Z (MUX2D1BWP16P90LVT)                       0.01       0.08 r
counter/U15/ZN (NR2D1BWP16P90LVT)                       0.00       0.08 f
counter/count_reg[0]/D (DFQD2BWP16P90LVT)               0.00       0.08 f
data arrival time                                                  0.08

clock clk (rise edge)                                   0.05       0.05
clock network delay (ideal)                             0.00       0.05
counter/count_reg[0]/CP (DFQD2BWP16P90LVT)              0.00       0.05 r
library hold time                                       0.01       0.05
data required time                                                 0.05
-----------------------------------------------------------------------
data required time                                                 0.05
data arrival time                                                 -0.08
-----------------------------------------------------------------------
slack (MET)                                                        0.03
```

17. In the Toolbox, select **Select → Path From/Through/To**, set the delay type to max, and check the critical path.



18. In the command line, enter: **change_names -hierarchy -rule verilog**

```
design_vision> change_names -hierarchy -rule verilog
1
```

19. In the command line, enter: **report_qor > traffic_light.qor** to save the report.

```
design_vision> report_qor > traffic_light.qor
```

20. In the Toolbox, select **File → Save As**, name the file **traffic_light_syn.v**, and click **Save** to store the synthesized file.
21. In the command line, enter: **write_sdf -version 1.0 -context verilog traffic_light_syn.sdf** to write the timing information into the SDF file.

```
design_vision> write_sdf -version 1.0 -context verilog traffic_light_syn.sdf
Information: Annotated 'cell' delays are assumed to include load delay. (UID-282)
Information: Writing timing information to file '/home/UGP/B123015021/HW4/traffic_light_syn.sdf'. (WT-3)
1
```

22. Modify the testbench by commenting out **`include "./traffic_light.v"**, then adding **`include "./traffic_light_syn.v"** and **`include "./N16ADFP_StdCell.v"**.

```
1 `timescale 1ns/10ps
2 // `include "./traffic_light.v"
3 `include "./traffic_light_syn.v"
4 `include "./N16ADFP_StdCell.v"
```

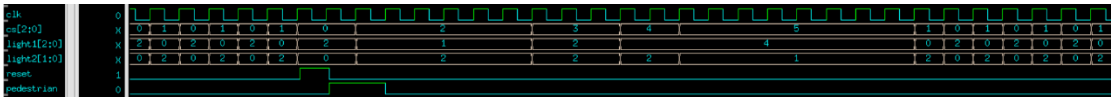
Inside the module, insert **initial $sdf_annotate ("traffic_light_syn.sdf", DUT);** to apply timing information from the SDF file to the design under test, ensuring that the synthesized design and standard cell library are correctly included for post-synthesis simulation.

```
1 traffic_light DUT (clk, reset, pedestrian, light1, light2);
2 initial $sdf_annotate ("traffic_light_syn.sdf", DUT);
```

23. In the Xshell terminal, enter **ncverilog tb_traffic_light.v +access+r** to rerun the simulation.



24. In Notepad++, check that the message **"Annotation completed successfully"** is displayed.
25. In the Xshell terminal, enter **nWave &** to launch nWave, open tb, and select the signals you want to observe.



➢ Synthesis Comparison

1. Clock Period

This section compares the differences between a large period and a small period, where the larger period is chosen to be 80 and the smaller period is chosen to be 10.

	Period 10	Period 80

<table>
<tr><td rowspan="3">Area</td><td>

```
****************************************
Report : area
Design : traffic_light
Version: Q-2019.12
Date   : Tue Nov 25 01:11:12 2025
****************************************

Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad

Number of ports:                           15
Number of nets:                            66
Number of cells:                           57
Number of combinational cells:             50
Number of sequential cells:                 6
Number of macros/black boxes:               0
Number of buf/inv:                         15
Number of references:                      18

Combinational area:                12.597120
Buf/Inv area:                       2.332800
Noncombinational area:              5.598720
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire l

Total cell area:                   18.195840
Total area:                 undefined

***** End Of Report *****
```

</td><td>

```
****************************************
Report : area
Design : traffic_light
Version: Q-2019.12
Date   : Tue Nov 25 01:11:12 2025
****************************************

Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad

Number of ports:                           15
Number of nets:                            66
Number of cells:                           57
Number of combinational cells:             50
Number of sequential cells:                 6
Number of macros/black boxes:               0
Number of buf/inv:                         15
Number of references:                      18

Combinational area:                12.597120
Buf/Inv area:                       2.332800
Noncombinational area:              5.598720
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire l

Total cell area:                   18.195840
Total area:                 undefined

***** End Of Report *****
```

</td></tr>
<tr><td>18.195840 μm^2</td><td>18.195840 μm^2</td></tr>
<tr><td>Same</td><td>Same</td></tr>
<tr><td rowspan="3">Power</td><td>

```
  Cell Internal Power  =   2.6015 uW   (82%)
  Net Switching Power  = 562.1918 nW   (18%)
                         ---------
Total Dynamic Power    =   3.1637 uW  (100%)

Cell Leakage Power     =  26.3927 uW


                 Internal         Switching
Power Group      Power            Power
--------------------------------------------------
io_pad             0.0000           0.0000
memory             0.0000           0.0000
black_box          0.0000           0.0000
clock_network      0.0000           0.0000
register       2.1600e-03       2.1771e-04
sequential         0.0000           0.0000
combinational  4.4148e-04       3.4448e-04
--------------------------------------------------
Total          2.6015e-03 mW    5.6219e-04 mW

***** End Of Report *****
```

</td><td>

```
  Cell Internal Power  = 325.1913 nW   (82%)
  Net Switching Power  =  70.2740 nW   (18%)
                         ---------
Total Dynamic Power    = 395.4652 nW  (100%)

Cell Leakage Power     =  26.3927 uW


                 Internal         Switching
Power Group      Power            Power
--------------------------------------------------
io_pad             0.0000           0.0000
memory             0.0000           0.0000
black_box          0.0000           0.0000
clock_network      0.0000           0.0000
register       2.7001e-04       2.7214e-05
sequential         0.0000           0.0000
combinational  5.5185e-05       4.3060e-05
--------------------------------------------------
Total          3.2519e-04 mW    7.0274e-05 mW

***** End Of Report *****
```

</td></tr>
<tr><td>3.1637 μW</td><td>395.4652 nW</td></tr>
<tr><td>Higher</td><td>Lower</td></tr>
<tr><td rowspan="3">Timing</td><td>

```
Point                                                   Incr       Path
------------------------------------------------------------------------
clock CLK_0 (rise edge)                                 5.00       5.00
clock network delay (ideal)                             0.00       5.00
cs_reg[2]/CP (DFQD2BWP16P90LVT)                         0.00       5.00 r
cs_reg[2]/Q (DFQD2BWP16P90LVT)                          0.03       5.03 f
U57/ZN (CKND1BWP16P90LVT)                               0.01       5.04 r
U60/ZN (NR2D1BWP16P90LVT)                               0.01       5.05 f
U49/ZN (ND2D1BWP16P90LVT)                               0.01       5.05 r
U66/ZN (OAI211D1BWP16P90LVT)                            0.01       5.06 f
U65/ZN (CKND1BWP16P90LVT)                               0.01       5.07 r
U44/ZN (OAI211D1BWP16P90LVT)                            0.01       5.08 f
counter/clear (counter)                                 0.00       5.08 f
counter/U3/ZN (CKND1BWP16P90LVT)                        0.01       5.08 r
counter/U6/ZN (ND3D1BWP16P90LVT)                        0.01       5.10 f
counter/U5/ZN (CKND1BWP16P90LVT)                        0.01       5.10 r
counter/U11/ZN (ND2D1BWP16P90LVT)                       0.01       5.11 f
counter/U7/ZN (OAI32D1BWP16P90LVT)                      0.01       5.12 r
counter/count_reg[2]/D (DFQD2BWP16P90LVT)               0.00       5.12 r
data arrival time                                                  5.12

clock CLK_0 (rise edge)                                15.00      15.00
clock network delay (ideal)                             0.00      15.00
counter/count_reg[2]/CP (DFQD2BWP16P90LVT)              0.00      15.00 r
library setup time                                     -0.01      14.99
data required time                                                14.99
------------------------------------------------------------------------
data required time                                                14.99
data arrival time                                                 -5.12
------------------------------------------------------------------------
slack (MET)                                                        9.86
```

</td><td>

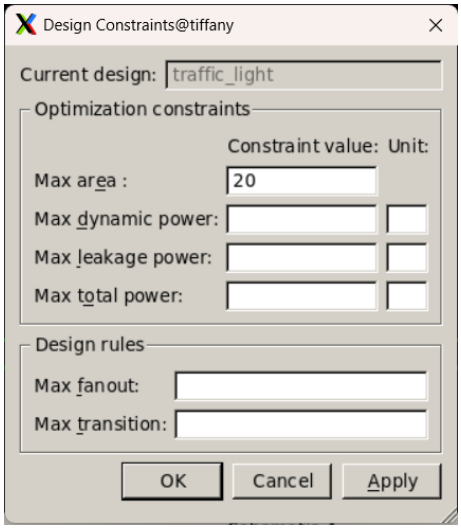
```
Point                                                   Incr       Path
------------------------------------------------------------------------
clock CLK_0 (rise edge)                                40.00      40.00
clock network delay (ideal)                             0.00      40.00
cs_reg[2]/CP (DFQD2BWP16P90LVT)                         0.00      40.00 r
cs_reg[2]/Q (DFQD2BWP16P90LVT)                          0.03      40.03 f
U61/ZN (CKND1BWP16P90LVT)                               0.01      40.04 r
U64/ZN (NR2D1BWP16P90LVT)                               0.01      40.05 f
U49/ZN (ND2D1BWP16P90LVT)                               0.01      40.05 r
U53/ZN (OAI211D1BWP16P90LVT)                            0.01      40.06 f
U52/ZN (CKND1BWP16P90LVT)                               0.01      40.07 r
U45/ZN (OAI211D1BWP16P90LVT)                            0.01      40.08 f
counter/clear (counter)                                 0.00      40.08 f
counter/U4/ZN (CKND1BWP16P90LVT)                        0.01      40.08 r
counter/U6/ZN (ND3D1BWP16P90LVT)                        0.01      40.10 f
counter/U5/ZN (CKND1BWP16P90LVT)                        0.01      40.10 r
counter/U11/ZN (ND2D1BWP16P90LVT)                       0.01      40.11 f
counter/U7/ZN (OAI32D1BWP16P90LVT)                      0.01      40.12 r
counter/count_reg[2]/D (DFQD2BWP16P90LVT)               0.00      40.12 r
data arrival time                                                 40.12

clock CLK_0 (rise edge)                               120.00     120.00
clock network delay (ideal)                             0.00     120.00
counter/count_reg[2]/CP (DFQD2BWP16P90LVT)              0.00     120.00 r
library setup time                                     -0.01     119.99
data required time                                               119.99
------------------------------------------------------------------------
data required time                                               119.99
data arrival time                                                -40.12
------------------------------------------------------------------------
slack (MET)                                                       79.86
```

</td></tr>
<tr><td>Slack = 9.86 (MET)</td><td>Slack = 79.86 (MET)</td></tr>
<tr><td>Lower Delay</td><td>Higher Delay</td></tr>
</table>

With a period of 80, the slack becomes significantly large. Although there is essentially no reduction in area, the decrease in power consumption is clearly noticeable.

2. Area Constraint



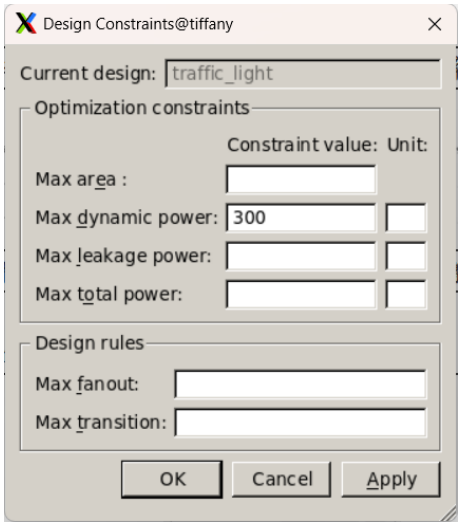
The objective is to reduce the Max Area constraint slightly, while keeping the Max Dynamic Power and Max Leakage Power constraints relatively relaxed.

	With Constraint	Without Constraint
Area	** Report : area Design : traffic_light Version: Q-2019.12 Date : Tue Nov 25 21:05:26 2025 ** Information: Updating design information... (U Library(s) Used: N16ADFP_StdCellff0p88v125c_ccs (File: /cad Number of ports: 15 Number of nets: 71 Number of cells: 62 Number of combinational cells: 55 Number of sequential cells: 6 Number of macros/black boxes: 0 Number of buf/inv: 14 Number of references: 25 Combinational area: 14.878080 Buf/Inv area: 2.384640 Noncombinational area: 5.598720 Macro/Black Box area: 0.000000 Net Interconnect area: undefined (Wire l Total cell area: 20.476800 Total area: undefined ***** End Of Report *****	** Report : area Design : traffic_light Version: Q-2019.12 Date : Tue Nov 25 20:41:52 2025 ** Information: Updating design information... (U Library(s) Used: N16ADFP_StdCellff0p88v125c_ccs (File: /cad Number of ports: 15 Number of nets: 75 Number of cells: 66 Number of combinational cells: 59 Number of sequential cells: 6 Number of macros/black boxes: 0 Number of buf/inv: 18 Number of references: 25 Combinational area: 15.552000 Buf/Inv area: 3.006720 Noncombinational area: 5.598720 Macro/Black Box area: 0.000000 Net Interconnect area: undefined (Wire l Total cell area: 21.150720 Total area: undefined ***** End Of Report *****

	$20.476800\ \mu m^2$	$21.150720\ \mu m^2$
	Smaller	Larger
Power	Cell Internal Power = 279.2013 uW (82%) Net Switching Power = 59.6019 uW (18%) Total Dynamic Power = 338.8032 uW (100%) Cell Leakage Power = 33.1469 uW Power Group / Internal Power / Switching Power io_pad 0.0000 0.0000 memory 0.0000 0.0000 black_box 0.0000 0.0000 clock_network 0.0000 0.0000 register 0.2155 2.0262e-02 sequential 0.0000 0.0000 combinational 6.3749e-02 3.9340e-02 Total 0.2792 mW 5.9602e-02 mW ***** End Of Report *****	Cell Internal Power = 276.7772 uW (82%) Net Switching Power = 62.2636 uW (18%) Total Dynamic Power = 339.0408 uW (100%) Cell Leakage Power = 32.5602 uW Power Group / Internal Power / Switching Power io_pad 0.0000 0.0000 memory 0.0000 0.0000 black_box 0.0000 0.0000 clock_network 0.0000 0.0000 register 0.2155 1.9208e-02 sequential 0.0000 0.0000 combinational 6.1261e-02 4.3056e-02 Total 0.2768 mW 6.2264e-02 mW ***** End Of Report *****
	$338.8032\ \mu W$	$339.0404\ \mu W$
	Lower	Higher
Timing	Point / Incr / Path clock clk (rise edge) 0.05 0.05 clock network delay (ideal) 0.00 0.05 cs_reg[1]/CP (DFQD2BWP16P90LVT) 0.00 0.05 r cs_reg[1]/Q (DFQD2BWP16P90LVT) 0.03 0.08 f U43/ZN (CKND2BWP16P90LVT) 0.01 0.09 r U50/Z (AN3D1BWP16P90LVT) 0.01 0.10 r U67/Z (MUX2D1BWP16P90LVT) 0.01 0.11 r U68/Z (MUX2D1BWP16P90LVT) 0.01 0.12 r U69/ZN (IOA21D1BWP16P90LVT) 0.01 0.13 r U51/ZN (IAO21D1BWP16P90LVT) 0.01 0.14 r cs_reg[0]/D (DFQD2BWP16P90LVT) 0.00 0.14 r data arrival time 0.14 clock clk (rise edge) 0.15 0.15 clock network delay (ideal) 0.00 0.15 cs_reg[0]/CP (DFQD2BWP16P90LVT) 0.00 0.15 r library setup time -0.01 0.14 data required time 0.14 data required time 0.14 data arrival time -0.14 slack (MET) 0.00	Point / Incr / Path clock clk (rise edge) 0.05 0.05 clock network delay (ideal) 0.00 0.05 cs_reg[1]/CP (DFQD2BWP16P90LVT) 0.00 0.05 r cs_reg[1]/Q (DFQD2BWP16P90LVT) 0.03 0.08 f U43/ZN (CKND2BWP16P90LVT) 0.01 0.09 r U51/Z (AN3D1BWP16P90LVT) 0.01 0.10 r U68/Z (MUX2D1BWP16P90LVT) 0.01 0.11 r U69/Z (MUX2D1BWP16P90LVT) 0.01 0.12 r U70/ZN (IOA21D1BWP16P90LVT) 0.01 0.13 r U52/ZN (IAO21D1BWP16P90LVT) 0.01 0.14 r cs_reg[0]/D (DFQD2BWP16P90LVT) 0.00 0.14 r data arrival time 0.14 clock clk (rise edge) 0.15 0.15 clock network delay (ideal) 0.00 0.15 cs_reg[0]/CP (DFQD2BWP16P90LVT) 0.00 0.15 r library setup time -0.01 0.14 data required time 0.14 data required time 0.14 data arrival time -0.14 slack (MET) 0.00
	Slack = 0.00 (MET)	Slack = 0.00 (MET)
	Same	Same

The above comparison between runs with and without an Area Constraint was conducted under the same condition, with the period fixed at 0.1.
We can observe that although the synthesized area slightly exceeds my set target of 20, it is still very close. What is surprising is that the power consumption is also lower, which was unexpected.

3. Power Constraint



The objective is to reduce the Max Dynamic Power constraint slightly, while keeping the Max Area and Max Leakage Power constraints relatively relaxed.

	With Constraint	Without Constraint
Area	``` ** Report : area Design : traffic_light Version: Q-2019.12 Date : Tue Nov 25 21:22:42 2025 ** Information: Updating design information... (U Library(s) Used: N16ADFP_StdCellff0p88v125c_ccs (File: /cad Number of ports: 15 Number of nets: 78 Number of cells: 69 Number of combinational cells: 62 Number of sequential cells: 6 Number of macros/black boxes: 0 Number of buf/inv: 21 Number of references: 23 Combinational area: 15.811200 Buf/Inv area: 3.473280 Noncombinational area: 5.598720 Macro/Black Box area: 0.000000 Net Interconnect area: undefined (Wire l Total cell area: 21.409920 Total area: undefined ***** End Of Report ***** ```	``` ** Report : area Design : traffic_light Version: Q-2019.12 Date : Tue Nov 25 20:41:52 2025 ** Information: Updating design information... (U Library(s) Used: N16ADFP_StdCellff0p88v125c_ccs (File: /cad Number of ports: 15 Number of nets: 75 Number of cells: 66 Number of combinational cells: 59 Number of sequential cells: 6 Number of macros/black boxes: 0 Number of buf/inv: 18 Number of references: 25 Combinational area: 15.552000 Buf/Inv area: 3.006720 Noncombinational area: 5.598720 Macro/Black Box area: 0.000000 Net Interconnect area: undefined (Wire l Total cell area: 21.150720 Total area: undefined ***** End Of Report ***** ```
	$21.409920\ \mu m^2$	$21.150720\ \mu m^2$
	Larger	Smaller

<table>
<tr><td rowspan="3">Power</td><td>

```
  Cell Internal Power  = 276.1554 uW   (81%)
  Net Switching Power  =  62.8427 uW   (19%)
                         ---------
Total Dynamic Power    = 338.9981 uW  (100%)

Cell Leakage Power     =  33.1568 uW


                   Internal         Switching
Power Group        Power            Power
--------------------------------------------------
io_pad             0.0000           0.0000
memory             0.0000           0.0000
black_box          0.0000           0.0000
clock_network      0.0000           0.0000
register           0.2155           1.9217e-02
sequential         0.0000           0.0000
combinational      6.0659e-02       4.3626e-02
--------------------------------------------------
Total              0.2762 mW        6.2843e-02 mW

***** End Of Report *****
```

</td><td>

```
  Cell Internal Power  = 276.7772 uW   (82%)
  Net Switching Power  =  62.2636 uW   (18%)
                         ---------
Total Dynamic Power    = 339.0408 uW  (100%)

Cell Leakage Power     =  32.5602 uW


                   Internal         Switching
Power Group        Power            Power
--------------------------------------------------
io_pad             0.0000           0.0000
memory             0.0000           0.0000
black_box          0.0000           0.0000
clock_network      0.0000           0.0000
register           0.2155           1.9208e-02
sequential         0.0000           0.0000
combinational      6.1261e-02       4.3056e-02
--------------------------------------------------
Total              0.2768 mW        6.2264e-02 mW

***** End Of Report *****
```

</td></tr>
<tr><td>$338.9981\ \mu W$</td><td>$339.0404\ \mu W$</td></tr>
<tr><td>Lower</td><td>Higher</td></tr>
<tr><td rowspan="3">Timing</td><td>

```
Point                                   Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                   0.05       0.05
clock network delay (ideal)             0.00       0.05
cs_reg[1]/CP (DFQD2BWP16P90LVT)         0.00       0.05 r
cs_reg[1]/Q (DFQD2BWP16P90LVT)          0.03       0.08 f
U42/ZN (INVD2BWP16P90LVT)               0.01       0.09 r
U49/Z (AN3D1BWP16P90LVT)                0.01       0.10 r
U70/Z (MUX2D1BWP16P90LVT)               0.01       0.11 r
U71/Z (MUX2D1BWP16P90LVT)               0.01       0.12 r
U72/ZN (IOA21D1BWP16P90LVT)             0.01       0.13 r
U73/Z (OA21D1BWP16P90LVT)               0.01       0.14 r
cs_reg[0]/D (DFQD2BWP16P90LVT)          0.00       0.14 r
data arrival time                                  0.14

clock clk (rise edge)                   0.15       0.15
clock network delay (ideal)             0.00       0.15
cs_reg[0]/CP (DFQD2BWP16P90LVT)         0.00       0.15 r
library setup time                     -0.01       0.14
data required time                                 0.14
-----------------------------------------------------------
data required time                                 0.14
data arrival time                                 -0.14
-----------------------------------------------------------
slack (MET)                                        0.00
```

</td><td>

```
Point                                   Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                   0.05       0.05
clock network delay (ideal)             0.00       0.05
cs_reg[1]/CP (DFQD2BWP16P90LVT)         0.00       0.05 r
cs_reg[1]/Q (DFQD2BWP16P90LVT)          0.03       0.08 f
U43/ZN (CKND2BWP16P90LVT)               0.01       0.09 r
U51/Z (AN3D1BWP16P90LVT)                0.01       0.10 r
U68/Z (MUX2D1BWP16P90LVT)               0.01       0.11 r
U69/Z (MUX2D1BWP16P90LVT)               0.01       0.12 r
U70/ZN (IOA21D1BWP16P90LVT)             0.01       0.13 r
U52/ZN (IAO21D1BWP16P90LVT)             0.01       0.14 r
cs_reg[0]/D (DFQD2BWP16P90LVT)          0.00       0.14 r
data arrival time                                  0.14

clock clk (rise edge)                   0.15       0.15
clock network delay (ideal)             0.00       0.15
cs_reg[0]/CP (DFQD2BWP16P90LVT)         0.00       0.15 r
library setup time                     -0.01       0.14
data required time                                 0.14
-----------------------------------------------------------
data required time                                 0.14
data arrival time                                 -0.14
-----------------------------------------------------------
slack (MET)                                        0.00
```

</td></tr>
<tr><td>Slack = 0.00 (MET)</td><td>Slack = 0.00 (MET)</td></tr>
<tr><td>Same</td><td>Same</td></tr>
</table>

It can be observed that the synthesized power is much higher than my target value of 300, and is even close to the result from the run without any constraints. A possible reason is that under the period of 0.1, the power has already been pushed close to its limit.

➢ Circuit Analysis

In the ADFP process, the area of a single NAND gate is 0.20736 μm². The estimated number of NAND gates can be obtained by dividing the total area by 0.20736. The logic depth can be determined from the critical path by counting how many gate-delay layers it passes through. In this way, we can compare the differences before and after synthesis.

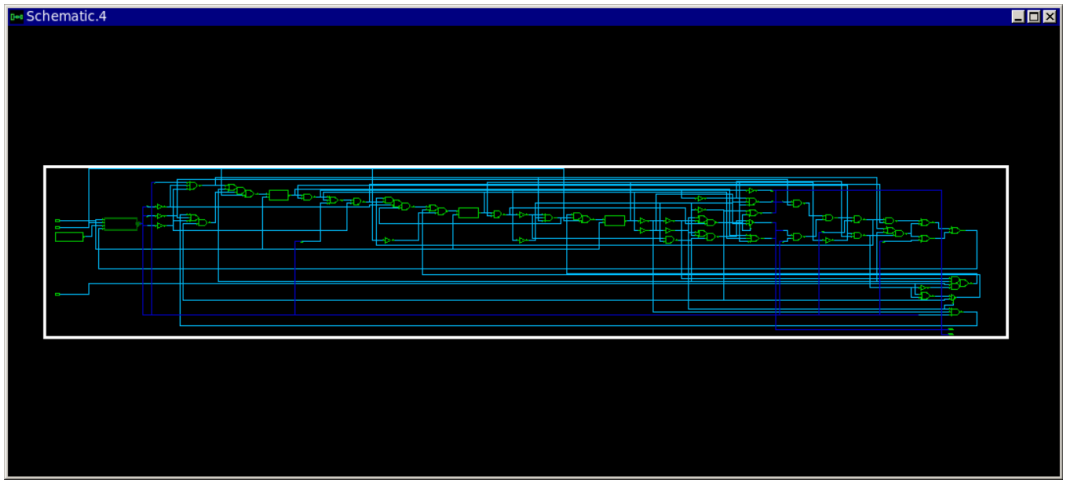
1. Pre-Synthesis

Since the design is written at the behavioral level, the following values are estimated based on my assumptions about how the circuit is likely structured.

- Total Number of Gates: 145
- Logic Levels: 7
- Critical Path Delay: 11Δ

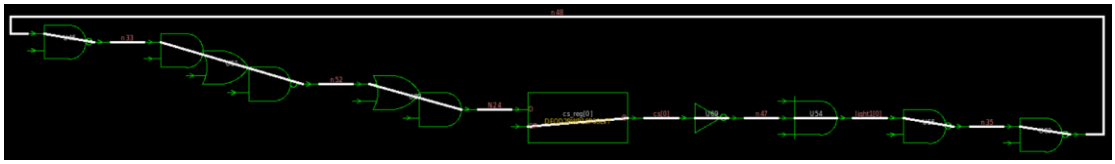
2. Post-Synthesis

i. Schematic View



- Total Number of Gates: $\frac{22.913281}{0.20736} \approx 110.5$

ii. Critical Path



- Logic Levels: 12
- Critical Path Delay: 25Δ

3. Comparison

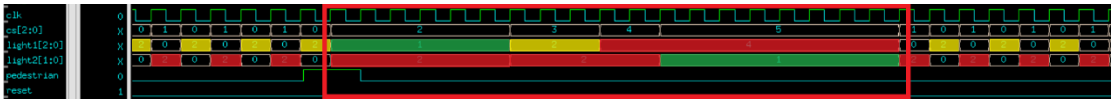
	Pre-Synthesis	Post-Synthesis
Number of Gates	145	110.5
Logic Levels	7	12
Critical Path	11Δ	25Δ

From the results, we can observe that the number of logic gates after synthesis is significantly smaller than before synthesis, but the logic depth and delay both increased. Although the synthesized data are as listed above, the results may also vary depending on factors such as the applied constraints or the input/output delay settings.
The synthesis data presented here were obtained under the condition that the period was set to 0.09.

➢ Result Analysis

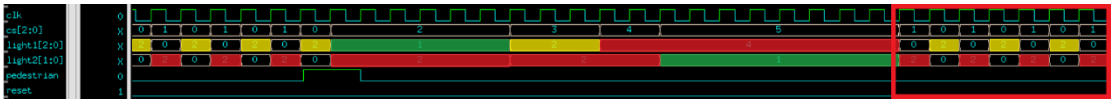


From the waveform, we verify that the system remains in its default state prior to any pedestrian activation. In this stage, Light1 consistently shows a blinking-yellow pattern, while Light2 outputs a blinking-red pattern. This behavior aligns with the design specification and confirms that the default caution mode is correctly implemented.



After the pedestrian button is activated, the waveform confirms that the system exits the default caution mode and proceeds through the multi-stage sequence. Light1 first switches to green while Light2 remains red for 6 clock cycles, then transitions to a yellow indication for 3 cycles,

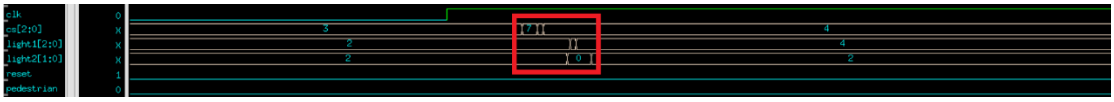
followed by a brief all-red phase lasting 2 cycles. The system then enters the road-crossing stage, during which Light1 stays red and Light2 turns green for 8 cycles. All observed output patterns and interval durations match the intended design, verifying correct operation during the pedestrian-handling process.



After completing the final stage, the waveform shows that the controller transitions cleanly back into its default blinking mode. At this point, Light1 resumes its blinking-yellow pattern and Light2 returns to its blinking-red pattern, confirming that the system correctly re-enters the default state once the full sequence has concluded.

➢ Delay Effect

After synthesis, the waveform occasionally exhibits brief unknown values at the moment the inputs change. These transient unknowns typically arise from gate-level propagation delays, which causes input arrival mismatches during signal transitions.



To avoid accidentally treating these harmless glitches as errors during comparison with the golden model, I added a #1 wait before checking the outputs in each cycle. This gives the circuit time to settle, ensuring that the verification is based only on the final stable result.

➢ Testbench Design

The testbench is designed to validate the traffic-light controller by comparing its output against a golden model. It begins by applying a reset signal to ensure that the FSM enters a known initial state. The core of the checking mechanism is the expected_lights() function, which serves as a reference model. The testbench reads the DUT's internal state and uses that value to determine the correct output pattern for {Light1, Light2}.

The testbench performs output comparisons in two phases. First, it verifies the controller's default behavior with no pedestrian input, confirming that the blinking-mode outputs are correct. It then activates the pedestrian signal and checks whether the controller advances through each stage of the required sequence with the correct output patterns and interval durations. If any mismatch happens between the DUT outputs and the golden model, the error counter increments and reports detailed failing information.

At the end of the simulation, the final error count determines whether the test passed or failed. If no errors are detected, the testbench prints an "ALL PASS" banner to indicate that the implementation behaves correctly under all conditions.

➢ Discussion & Reflection

In some designs, the counter continuously increments throughout all states and resets only after completing the entire sequence, but this approach requires a wide counter and leads to higher hardware cost. In contrast, the counter in my design uses only 3 bits, since it depends solely on the longest duration among all states, and it is shared across every FSM cycle. This significantly reduces the overall area. The trade-off is that the counter's reset signal becomes more critical and must be handled carefully to ensure correct operation.

When applying different clock periods, I observed that both power consumption and data-arrival times can vary dramatically. As I tightened

the clock period to push the slack closer to zero, the power consumption increased significantly, reaching approximately $350\ \mu W$. In contrast, when a much larger period was applied, the power consumption dropped to the nanowatt level, and the resulting slack became extremely large. Despite these large changes in timing and power, the synthesized area shows only minimal change. This shows that the clock period primarily influences power and timing behavior, while having little impact on area in my traffic light design.

Designing this sequential circuit showed me how important precise timing and signal control are in digital systems. Since this was my first time implementing both an FSM and a synchronous counter, I realized that coordinating the two is not easy and requires careful reset handling, timing alignment, and a clear design flow. Even small mistakes in control signals or timing assumptions can lead to incorrect state transitions or counter behavior. Through this homework, I gained a much stronger understanding of how to design a sequential circuit through proper state-switching logic, and I am looking forward to taking on more complex designs in the future.